

BAR, Montenegro

WMO: 134610

Lat: 42.100N

Lon: 19.100E

Elev: 6

StdP: 101.25

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.6	3.2	-14.2	1.1	4.0	-11.3	1.4	5.4	13.3	1.2	11.1	3.9	5.1	50	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	32.1	24.4	31.2	24.3	30.4	23.9	26.4	30.9	25.7	30.3	24.9	29.6	3.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.1	30.4	24.1	19.1	29.7	23.3	18.1	29.1	82.3	31.1	79.0	30.2	75.8	29.7	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.1	7.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	9.5	10.6	12.5	15.8	19.3	23.6	25.9	26.2	23.0	18.8	15.2	11.4		
	DBStd	6.29	3.23	2.99	2.46	2.00	2.53	2.48	1.81	1.69	2.30	2.21	2.92	3.16		
	HDD10.0	101	46	23	7	0	0	0	0	0	0	0	3	22		
	HDD18.3	1107	273	217	180	81	18	1	0	0	1	21	100	216		
	CDD10.0	2903	31	41	86	174	288	408	492	501	389	273	157	65		
	CDD18.3	867	0	1	0	4	48	158	233	243	139	36	5	0		
	CDH23.3	7492	0	3	0	11	222	1327	2381	2495	947	100	7	0		
(13)	CDH26.7	2347	0	0	0	0	19	354	846	917	209	1	1	0		
(14)	Wind	WSAvg	3.5	4.0	4.0	3.9	3.3	3.2	3.2	3.1	3.4	3.4	3.5	3.9		
(15) Precipitation	PrecAvg	1402	148	134	126	129	77	59	30	45	139	153	196	174		
	PrecMax	1957	351	335	361	294	168	232	99	190	359	252	431	343		
	PrecMin	1013	26	15	0	18	11	6	0	0	11	5	70	0		
	PrecStd	276	94	70	84	65	43	51	27	51	94	70	73	90		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	21.4	21.0	24.9	28.7	32.1	33.3	33.6	30.7	26.4	23.7	19.9		
		MCWB	11.7	13.0	12.4	17.4	22.5	23.2	24.5	23.9	23.8	20.3	17.0	15.1		
	2%	DB	16.3	18.5	19.3	22.7	26.8	30.8	32.1	32.3	29.8	25.3	21.9	18.1		
		MCWB	13.0	12.2	13.1	16.0	20.5	23.8	24.7	24.6	23.3	20.0	17.2	14.3		
	5%	DB	15.4	16.3	18.1	21.3	25.6	29.9	31.2	31.4	28.9	24.5	21.0	17.1		
		MCWB	12.5	11.7	12.8	15.7	19.6	23.6	24.5	24.3	22.5	19.4	16.8	14.1		
	10%	DB	14.3	15.2	17.0	20.2	24.4	28.8	30.4	30.5	28.0	23.5	20.0	16.2		
		MCWB	11.3	11.5	12.4	15.1	19.0	22.7	24.1	23.8	21.8	18.5	15.8	12.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	15.7	15.6	19.0	23.6	26.3	27.6	27.6	25.7	22.4	19.8	16.9		
		MCDB	16.0	17.6	18.3	23.1	27.7	30.4	31.3	31.7	29.5	24.2	21.2	18.1		
	2%	WB	14.1	14.1	14.8	17.6	21.8	25.3	26.4	26.5	24.5	21.2	18.7	15.3		
		MCDB	15.3	15.9	17.3	21.3	26.1	29.8	30.9	31.2	28.9	24.3	21.0	17.2		
	5%	WB	12.7	13.3	14.1	16.7	20.6	24.3	25.7	25.7	23.4	20.2	17.6	14.5		
		MCDB	14.6	15.4	16.6	20.1	24.7	29.1	30.3	30.5	28.1	23.7	20.2	16.6		
	10%	WB	11.8	12.3	13.2	15.9	19.5	23.4	24.9	24.9	22.5	19.3	16.3	13.5		
		MCDB	13.9	14.9	15.8	19.1	23.5	28.0	29.7	29.6	27.2	22.6	19.0	15.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.9	7.2	7.9	8.5	8.6	8.9	9.0	8.6	8.4	7.1	6.6		
		MCDBR	6.7	7.7	8.4	9.4	9.4	9.1	9.4	9.3	9.2	9.8	7.9	7.1		
		MCWBR	4.7	4.9	6.1	6.6	6.8	7.1	7.5	7.4	7.2	6.9	5.2	5.2		
	5% WB	MCDBR	5.9	6.2	7.7	8.9	9.1	9.0	9.1	9.1	8.9	9.2	6.9	6.0		
		MCWBR	4.7	4.8	6.2	6.7	6.9	7.5	7.9	7.9	7.5	6.9	5.4	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.338	0.361	0.404	0.454	0.442	0.425	0.426	0.438	0.419	0.391	0.365	0.340		
	taud		2.445	2.367	2.260	2.139	2.242	2.335	2.325	2.284	2.331	2.386	2.412	2.446		
	Ebn at Noon		800	834	836	817	838	852	846	823	812	790	757	762		
	Edh at Noon		78	97	122	147	136	125	125	126	112	95	80	72		
(44) All-Sky Solar Radiation	RadAvg		1.69	2.44	3.67	4.84	6.03	6.92	7.13	6.30	4.54	3.12	1.96	1.51		
	RadStd		0.18	0.36	0.49	0.49	0.46	0.43	0.35	0.37	0.34	0.30	0.19	0.26		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-78	N/A	N/A

Nomenclature: See separate page